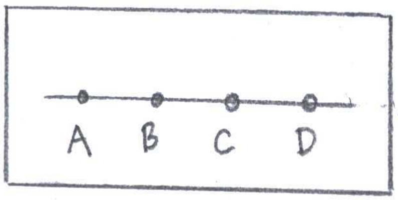
Euclidean and NonEuclidean Geometry:

Exercises on Betweenness:

1.



a) A, B, C, and D are four distinct points:

Proof:

Hypothesis	Evidence
1) From points B and D, other points reside on $\overleftrightarrow{BD}$, such as A and C.	Axiom B-2

b) A, B, C, and D are collinear:

Proof:

Hypothesis	Evidence
1) All points on line $\overleftrightarrow{BD}$ belong to segment BD on $\overleftrightarrow{BD}$	Euclids Postulate 1

c) (pg 110) "Corollary to Axiom B-4"

"For every line ℓ and for any three points A, B, and C not lying on ℓ

(i) If A and B are on the same side of ℓ and if B and

C are on the same side of ℓ, then A and C are on the same side of ℓ

(ii) If A and B are on opposite sides of ℓ and if B and C are on opposite sides of ℓ, then A and C are on the same side of ℓ.

(iii) If A and B are on same side of ℓ and B and C are on opposite sides of ℓ, then A and C are on opposite sides of ℓ."

Proof:

Hypothesis	Evidence
1) If B and C were on opposite sides of ℓ, then A and C are on the same side	Axiom B-4(ii)
2) A and C are on opposite sides	False RAA

2. a) (pg 109) "Proposition 3.1"

"For any two points A and B

(i) $\overrightarrow{AB} \cap \overrightarrow{BA} = AB$

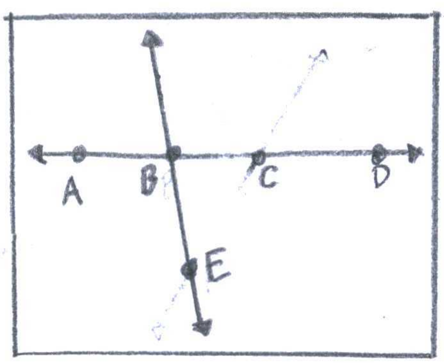
(ii) $\overrightarrow{AB} \cup \overrightarrow{BA} = \{\overleftrightarrow{AB}\}$"

Proof:

Hypothesis	Evidence
1) $AB \subset \overrightarrow{AB}$ and $AB \subset \overrightarrow{BA}$	(pg 16) "Definition - Segment" (pg 18) "Definition - Ray"
2) $AB \subset \overrightarrow{AB} \cap \overrightarrow{BA}$	(pg 45) "Definition - Intersection"
3) Point C belongs to Segment AB	RAA Premise
4) A, B, and C are collinear points, and	Axiom B-1
$A*C*B$, $A*B*C$ or $C*A*B$	Axiom B-3
5) i) Point C is on $\overrightarrow{AB}$ for $A*B*C$	(pg 16) "Definition - Ray"

ii) Point C is on $\overrightarrow{BA}$ for B*A*C	(pg 16) "Definition - Ray"
iii) Point C is between A and B for A*C*B	(pg 16) "Definition - Ray"
6) A*B*C, C*A*B and A*C*B are lines	Axiom B-1 (pg 12) "Definition - Lines"

b) (pg 113) "Proposition 3.3"



"A*B*D"

"Given A*B*C and A*C*D. Then B*C*D and A*B*D"

Proof

Hypothesis	Evidence
1) A, B, C, and D are four distinct collinear points	
2) Point E exists not on the line through A,B,C,D	Proposition 2.3

3) A line $\overleftrightarrow{EB}$ meets AD at point B. Also, A and D are on opposite sides of $\overleftrightarrow{EB}$	RAA hypothesis
4) C and D are on opposite sides	Contrary to RAA hypothesis
5) Then $\overrightarrow{EB}$ meets CD in a point between C and D	(pg 110) "Definition - opposite sides"
6) At point B	Proposition 2.1
7) Thus, C*B*D and B*C*D	RAA falsified because Axiom B-4
8) So, C and D are on the the same side	RAA conclusion
9) A and C are on opposite sides to $\overrightarrow{EB}$	Steps #3-8 and corollary to Axiom B-4

10) Point B is the intersection for the lines $\overleftrightarrow{EB}$ and $\overleftrightarrow{AD}$. Between A and C is point B.	(pg 110) "Definition - opposite sides" Proposition 2.1

c) <u>Converse to Proposition 3.3.</u>

Proof:

Hypothesis	Evidence
1) Point C is the intersection of $\overleftrightarrow{EC}$ and $\overleftrightarrow{BD}$. Point C is between B and D.	Proposition 2.1 (pg 110) "Definition - opposite sides"
2) B and D are on opposite sides of $\overleftrightarrow{EC}$	Hypothesis
3) Points A and B are on the same side of $\overleftrightarrow{EC}$	Hypothesis

4) $A*B*C$ and $A*C*B$	False RAA
5) The point for intersection is C	Proposition 2.1
6) $\overleftrightarrow{EC}$ meets $\overleftrightarrow{AB}$ in a point between A and B	(pg 110) "Definition - opposite sides"
7) A and B must be on opposite sides the line $\overleftrightarrow{EC}$	False RAA
8) Points A and B are on the same side	True conclusion from hypothesis, so $A*B*C$
9) A and D are on opposit sides line $\overleftrightarrow{EC}$	Steps #3-8
10) Point E coincides not on line through A, B, C, D	Proposition 2.3

10) A, B, C, and D are four distinct collinear points	

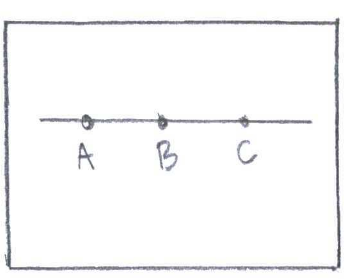
d) (pg 113) "Corollary to Proposition 3.3"

"Given $A*B*C$ and $B*C*D$. Then $A*B*D$ and $A*C*D$"

Proof:

Hypothesis	Evidence
1) Points B and C form a ray	(pg 12) "Definition-Line"
2) Both points A and D are on $\overrightarrow{BC}$ or $\overrightarrow{CB}$	Proposition 3.1
3) $A*B*D$ and $A*C*D$	Axiom B-1 and Axiom B-4

3.



a) $\underline{AB \subset AC}$

Proof:

Hypothesis	Evidence
1) A, B, and C are collinear points	Axiom B-1

2) B is between A and C	Euclid's Postulate #2

b) <u>$AC \subseteq AB \cup BC$</u>

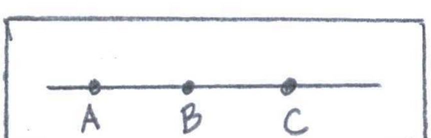
Proof: <u>Hypothesis</u>	Evidence
1) A, B, and C are three collinear points	Axiom B-1
2) $AB \cup BC$	
a) $AB \subseteq \overrightarrow{AB}$ and $BC \subseteq \overrightarrow{BC}$	(pg 16) "Definition - Rays"
	(pg 16) "Definition - segments"
b) $\overrightarrow{AB} \cup \overrightarrow{BC}$	(pg 45) "Definition - Union
3) $\overrightarrow{AB} \cup \overrightarrow{BC} = \overrightarrow{AC}$	(pg 45) "Definition - Union
4) $\overrightarrow{AC} \subset AC$	(pg 16) "Definition - Rays

c) (pg 114) "Proposition 3.5"

"Given $A * B * C$. Then $AC = AB \cup BC$ and B is the only point common to segments AB and BC"

Proof: Hypothesis	Evidence
1) A point P exists on A*B*C.	Axiom B-1
2) Point P is not B	RAA conjecture
3) So, point P is on segment AB or BC	
4) Two lines are possible: A*P*B and B*P*C	Axiom B-1
5) $AC = AB \cup BC$	
a) $A*P*B \subset A*B$	
b) $B*P*C \subset B*C$	
c) $AB \subset \overrightarrow{AB}$	"Definition - Ray"
d) $BC \subset \overrightarrow{BC}$	"Definition - Ray"
e) $\overrightarrow{AB} \cup \overrightarrow{BC}$	"Definition - Union"
f) $\overrightarrow{AC} = \overrightarrow{AB} \cup \overrightarrow{BC}$	
g) $\overrightarrow{AC} \not\subset AC$	"Definition -
6) $P = B$	RAA contradiction

4.



a) $\sim A*B*P \Rightarrow \sim A*C*P$

Proof: Hypothesis	Evidence

on $A*B*C$, then P is incident with all of them.	Collinear Proposition 3.3

b) $\overrightarrow{BA} \subset \overrightarrow{CA}$ and $\overrightarrow{BC} \subset \overrightarrow{AC}$ because A, B, and C are collinear and lay on the same line (Proposition 3.4).

c) B is the only point in common to $\overrightarrow{BA}$ and $\overrightarrow{BC}$:

Proof:

Hypothesis	Evidence.
1) Points A, B, C are collinear and union	Proposition 3.1
2) A point P is on $A*B*C$	Proposition 3.3
3) Point P is on $\overrightarrow{BA}$ or opposite ray $\overrightarrow{BC}$	Proposition 3.4
4) But $\overrightarrow{BA} \cup \overrightarrow{BC}$, so $P=B$	Proposition 3.5

5. (pg 115) "Proposition 3.6"

"Given $A*B*C$. Then B is the only

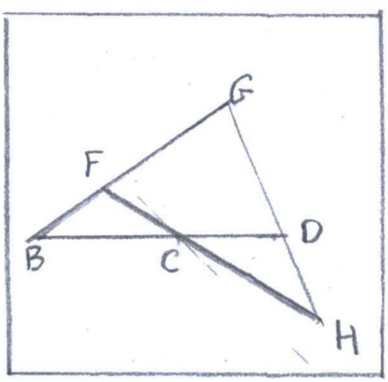
point in common

to rays $\overrightarrow{BA}$ and $\overrightarrow{BC}$, and $\overrightarrow{AB} = \overrightarrow{AC}$.

Proof:

Hypothesis	Evidence
1) Points A, B, C are collinear and union	Proposition 3.1
2) $\overrightarrow{AB} \subset \overrightarrow{AC}$	(pg 134) "Dedekinds Axiom"
3) If a point P is on $\overrightarrow{AB}$, then also $\overrightarrow{AC}$.	(pg 14) "Definition - Subset"
4) $\overrightarrow{AB} = \overrightarrow{AC}$	Outcome.

6.



Point C exists between B and D:

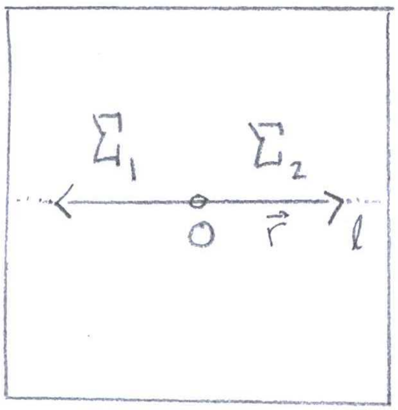
Proof:

Hypothesis	Evidence
1) Line BD is through B and D	Axiom I-1
2) Point F exists not on BD	Axiom I-3
3) Line BF through B and F.	Axiom I-1
4) Point G builds collinear B * F * G	(pg 70) "Definition - collinear"

5) Points B, F, and G are collinear	(pg 70) "Definition- collinear"
6) G and D are distinct points and D, B, and G are not collinear	Corallory to Axiom I-3
7) A point H exists such that G*D*H	(pg 70) "Definition- collinear"
8) Line $\overleftrightarrow{GH}$ exists	(pg) "Definition-line
9) H and F are distinct points	(pg) "Definition- Point"
10) Line $\overleftrightarrow{FH}$ exists	(pg) "Definition-Line"
11) D does not lie on $\overleftrightarrow{FH}$	Proposition 2.2
12) B does not lie on $\overleftrightarrow{FH}$	Proposition 2.2
13) G does not lie on $\overleftrightarrow{FH}$	Proposition 2.2
14) Points D, B, and G determine ΔDBG	(pg ???) "Definition- Triangle"

and $\overleftrightarrow{FH}$ intersects side BG in a point between B and G.	
15) H is the only point lying on both $\overleftrightarrow{FH}$ and $\overleftrightarrow{GH}$	Proposition 2.1
16) No point between G and D lies on $\overleftrightarrow{FH}$	Outcome steps #11/13
17) Hence, $\overleftrightarrow{FH}$ intersects side BD in a point between B and D.	Proposition 3.9
18) Point C is between D and B.	Proposition 3.5

7.



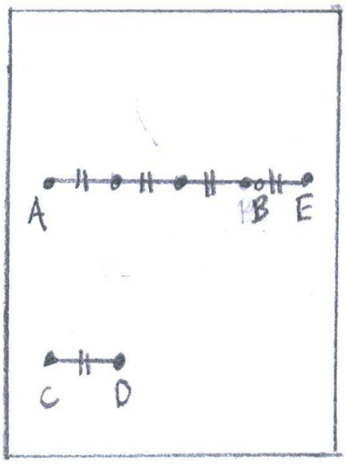
"Dedekind's Axiom"

a) **Dedekind's Axiom for Rays:**

"For line ℓ intersecting ray $\vec{r}$ at vertex A. Then a disjoint union $\Sigma_1 \sqcup \Sigma_2$ of two nonempty sets such each are exclusive subsets. Also, point O separates a set with A at O and the other is equal to the compliment.

Dedekind's Axiom for Segments:

"For line ℓ intersecting segment s at vertex A. Then a disjoint union $\Sigma_1 \sqcup \Sigma_2$ of two nonempty sets such each are exclusive subsets. Also, point O separates a set with A at O and the other is equal to the compliment."



"Archimedes Axiom"

b) (pg 132) "Archimedes Axiom"

"If CD is any segment, A any point, and ray r any ray with vertex A, then for every point $B \neq A$ on r there is a number n such that when CD is laid off n times on r starting at A, a point E is reached such that $n \cdot CD \cong AE$ and either $B = E$ or B is between A and E."

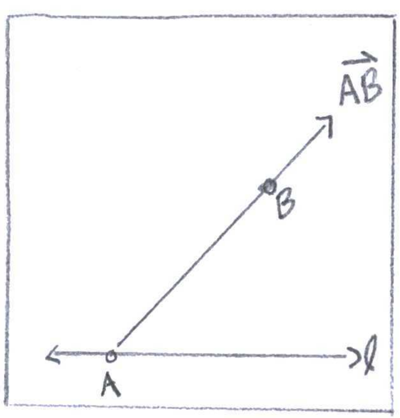
$$\exists \overline{AB}(\overline{AB} \in \ell) \,\&\, \exists A(A \in \ell) \longrightarrow \overline{\overline{AB}}(\overline{\overline{AB}} \in \ell)$$

<u>Informal proving about infinite points on a segment with a minimum five points:</u>

For a line ℓ, exists two distinct points incident with ℓ (Axiom I-1).

Another point exists between the two distinct points (Axiom B-2). The fourth point resides on ℓ between the new point and distinct point (Axiom B-2). Lastly, a fifth point between any two points (Axiom B-2). Essentially, an infinite application to Axiom B-2.

9.

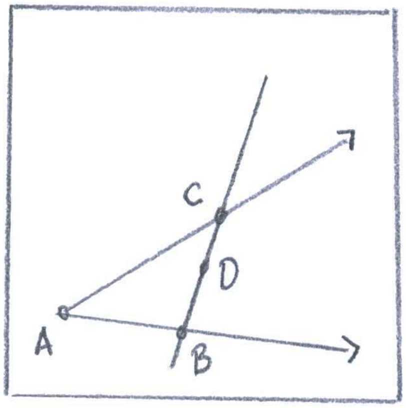


<u>Every part of ray AB is on the same side of ℓ:</u>

Hypothesis	Evidence
1) A point X is between A and B	Axiom B-1
2) A point Y is between A and X	Axiom B-1
3) Further, another point is between A and Y.	Axiom B-1

4) Points except A are on the same side of ℓ	Axiom B-4.

10.



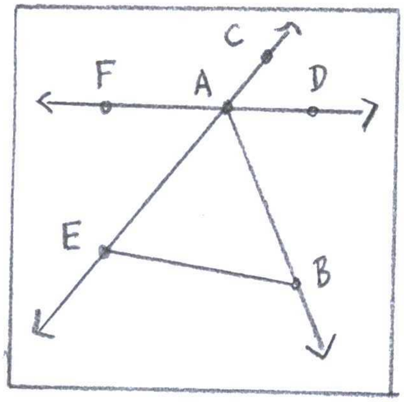
(pg 115) "Proposition 3.17"

"Given an angle $\measuredangle CAB$ and point D lying on line $\overleftrightarrow{BC}$. Then D is the interior of $\measuredangle CAB$ and only if $B * D * C$."

Proof:

Hypothesis	Evidence
1) From angle $\measuredangle CAB$, point D is on the same side $\overrightarrow{AC}$ as to $\overrightarrow{AB}$	(pg 115) "Definition-interior"
2) Points B, D, and C are collinear with a line incident all three points	(pg 70) "Definition-collinear"
3) $B * D * C$	Axiom B-1

11.



(pg 115) "Proposition 3.8"

"If D is in the interior of $\measuredangle CAB$, then a) so is every point on ray $\overrightarrow{AD}$ except A b) no point opposite $\overrightarrow{AD}$ is in the interior of $\measuredangle CAB$; and c) if $C * A * E$ then B is in the interior of $\measuredangle DAE$."

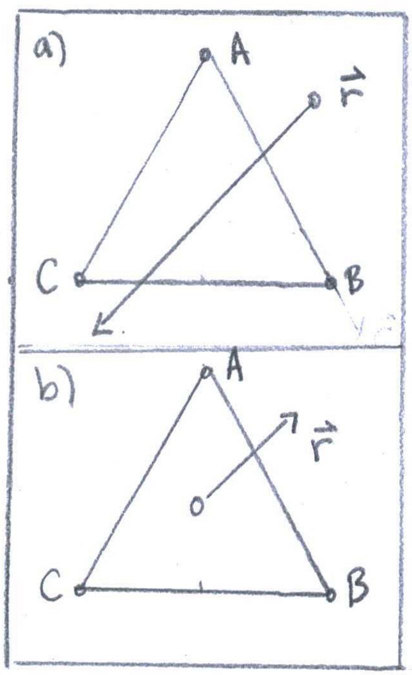
Proof:

Hypothesis	Evidence
1) Points on the same side of $\overleftrightarrow{FD}$ as D are opposite points on the other side $\overleftrightarrow{FD}$	Axiom B-4.
2) If a line has a interior ray, then also an exterior opposite.	(pg 13) "Definition-opposite"

3) $\measuredangle EAB$ is supplementary to $\measuredangle BAC$, so $\overrightarrow{AB}$ is between $\measuredangle CAD$.

(pg 19) "Definition- Supplementary Angles"

12.



"ray... emanating from exterior... or interior point about a triangle"

(pg 117) "Proposition 3.9"

(a) If a ray r emanating from an exterior point of ΔABC intersects side AB in a point between A and B, then r also intersects side AC or side BC.

(b) If a ray emanates from an interior point of ΔABC, then it intersects one of the sides, and if it does not pass through a vertex, it intersects only one side."

Proof:

Hypothesis	Evidence
1) If points A, B, C are noncollinear, then ray $\vec{r}$ intersects $\overline{AB}$ and, $\overline{AC}$ or $\overline{BC}$.	(pg 114) "Pasch's Theorem"
2) Else if, ray $\vec{r}$ is between $\overline{AC}$ and $\overline{BC}$, then $\vec{r}$ intersects only $\overline{AB}$	(pg 116) "Crossbar Theorem"

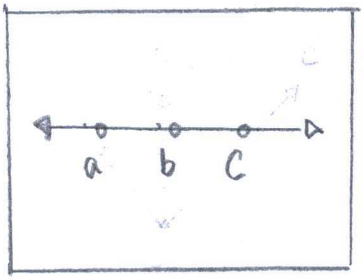
13. A line cannot be contained in the interior of a triangle:

Proof

Hypothesis	Evidence
1) A, B, and C are distinct noncollinear points	
2) Segments form between A, B, and C into AB, AC, and BC	(pg 16) "Definition - Segment"
3) A point appears inside the triangle. The point label - P.	
4) A line PQ is from point P to point Q.	(pg 12) "Definition - line"
5) The line PQ intersects a) Point A b) Point B c) Point C d) Segment AB e) Segment AC	Paschs Theorem

f) Segment BC

14.



"coterminal

a) Analogue to Axiom B-1:

"If $a*b*c$, then a, b, c are distinct and coterminal and $c*b*a$"

Analogue to Axiom B-2:

"If rays b and d are distinct, then rays a, c, and e reside on $\overleftrightarrow{bd}$ as distinct rays, coterminal, also $a*b*d$, $b*c*d$, and $b*d*e$."

Analogue to Axiom B-3:

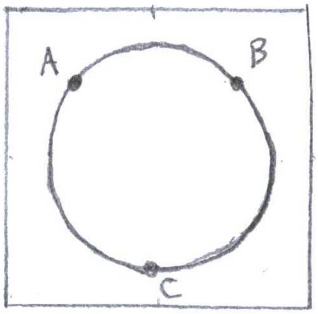
"If a, b, and c are on the same ray line, then one and only one ray is between the other two."

Analogue to Proposition 3.3:

"If $a*b*c$ and $a*c*d$, then $b*c*d$ and $a*b*d$."

The incorrect analogue parts are coterminal. Rays are between each other and not coterminal.

15.



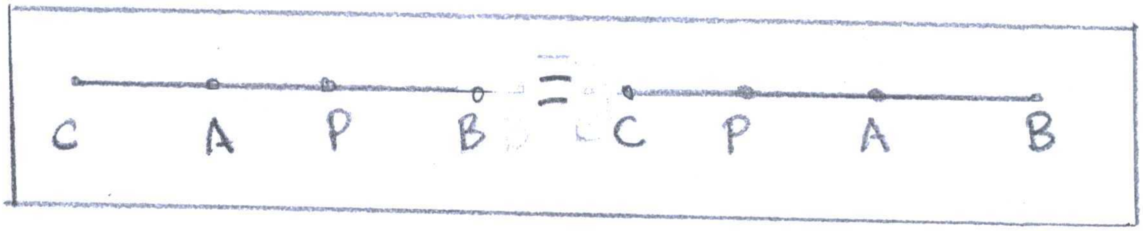
"three collinear points, no one of which is between the other two"

Betweeness. Axiom 3(pg 108) Fails interpretation in unclear circumstanes about lines. Definitions consider both circular and curved lines. So, in Axiom 3, "no one of which is between the other two" is necessary.

16.

Marvin Greenburg asked an "interpretation" to between. What about a line, such as C*A*B, with 'between' defining a segment from point to point. The line C*A*B is from the exercise. Axioms B-1,2,3 hold, but not Proposition 3.4 from the new definition -'between'. An example, point P is between A and B, in addition, the point A is between both P and B. The segment $\overline{AP}$

is irregardless to endpoints.
Between A and P means
between P and A in layout.



17. $\boxed{\dfrac{a}{2^n}}$

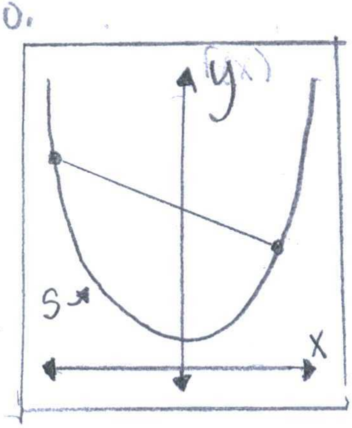
"Dyadic"

(pg 114) "Pasch's Theorem"

"If A, B, C are distinct noncollinear points and ℓ is any line intersecting AB in a point between A and B, then ℓ also intersects either AC and BC. If C does not lie on ℓ, then ℓ does not intersect both AC and BC."

The book presented an example on dyadic points. Vertices are at $(0,0)$, $(0,1)$, and $(\frac{1}{3},0)$. A line $3x+y=1$, supposedly never does

Scale in a dyadic plane.
If X-values scale by $n=0,1,2,3\ldots$
then the vertices too. The line.
models Pasch's theorem at $n=0$,
but not $n\geq 2$. Why? The line
becomes piecewise. The dyadic
function is $y=-\frac{3x}{2^n}+1$ and
requires a decent amount of
n-values.

10. 

(from exercise) "Convex"
"Whenever two points A and B are in S, the entire segment AB is contained in S"

Half-plane is convex:

Proof:

Hypothesis	Evidence
1) A half-plane has a line ℓ	Proposition 3.2
2) Set S for a half-plane is	$S=\{(x,y)\in\mathbb{R}^2\}$

the values greater or less than line ℓ	
3) If points A and B are greater than (x, y), then also segment AB	$S = \{(x,y) \in \mathbb{R}^2 : ax + by + c \geq 0\}$
4) If points A and B are less than (x, y), then also segment AB	$S = \{(x,y) \in \mathbb{R}^2 : ax + by + c \leq 0\}$

Interior of an angle is convex:

Proof: Hypothesis	Evidence
1) The set S of interior of an angle is less than 180° and greater than 0°	$S = \{(x \in \mathbb{R}) : 0 < x < 180\}$
2. Set S contains segment AB	

when points A and B are greater than 0° and less than 180°. By notation, point A is greater in degree to B.	$AB=\{A+tB \mid t\in[0,1], A>B\}\in S$

Interiors of a triangle are convex:

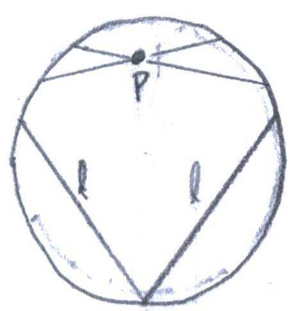
Proof:

Hypothesis	Evidence
1) The set S of interior angles is less than 180° and greater than 0°.	$S=\{x\in\mathbb{R}: 0<x<180\}$
2) Set S contains segment AB when points A and B less than 180° and greater than 0°. For notation A > B.	$AB=\{A+tB \mid t\in[0,1], A>B\}\in S$

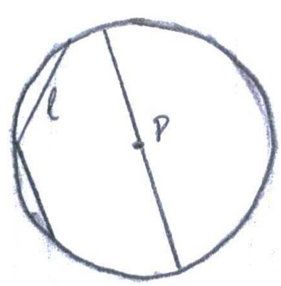
(pg 75) "Hyperbolic Parallel Property"

"For every line l and every point P not on l, there exist at least two lines through P parallel to l"

<u>Examples showing Hyperbolic Parallel Property in Chords around a Unit Circle:</u>



Author hearsays, "lines through point P are parallel to lines l."



Supposedly, the chord through P is parallel to lines l.

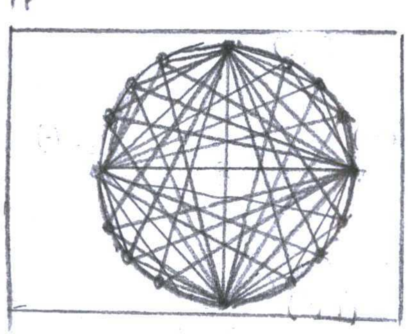
Exterior points of a triangle are not convex:

Proof:

Hypothesis	Evidence
1) Exterior, set S of a triangle is all points outside a triangle from 0 to infinity.	$S = \{(x,y) \in \mathbb{R}^2 :$ $x = (-\infty < (1-u-v)x_1 + ux_2 + vx_3 < \infty$ $y = (-\infty < (1-u-v)y_1 + uy_2 + vy_3 < \infty$ $u, v \geq 0, u+v \leq 1\}$ where $(x_1, y_1), (x_2 y_2)$ and (x_3, y_3) are vertices to triangle
2) If points A and B are outside the triangle, then not every segment AB.	$AB = \{A + tB \mid t \in [0,1], A > B\} \not\subseteq S.$

A triangle has finite points inside, so convex.

19.



"Chords on a Circle"

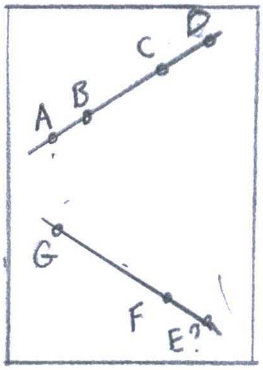
Exercises on Congruence:

20. (pg 124) "Proposition 3.11 (Segment Subtraction)"

Proof:

Hypothesis	Evidence
1) Assume on the contrary that BC is not congruent to EF	RAA Hypothesis
2) Then there is a point G on EF such that $BC \cong EG$	Euclid Postulate II
3) $G \neq F$	(pg 12) "Definition- point"
4) Since $AB \cong DE$, adding gives $AC \cong DG$	Axiom C-3
5) However, $AC \cong DF$	Contradiction
6) Hence, $DF \cong DG$	Requirement 0
7) Therefore, $F = G$	Reality
8) Our assumption led to a contradiction; hence, $BC \cong EF$	False RAA result

21. (pg 125) "Proposition 3.13a (Segment Ordering)"



"Exactly one of the following statements holds (trichotomy):

$AB < CD$, $AB \cong CD$ or $AB > CD$."

Proof

Hypothesis	Evidence
1) $AB \cong CD$	RAA Hypothesis #1
a) A point E resides on AB and F on CD, such $AE \cong CF$	Proposition 3.12
b) $EB \cong FD$	Proposition 3.11
2) $AB < CD$	RAA Hypothesis #2
a) A point G exists on CD, such $AB \cong CG$	Euclid Postulate II
b) $D \neq G$	(pg 12) "Definition - Point"
3) $AB > CD$	RAA Hypothesis #3
a) A point H exists on AB, such $AH \cong CD$	Euclid Postulate II
b) $B \neq H$	
4) If $D = G$ or $B = H$, then initial step #1.	False result

22

(pg 132) "Proposition 3.12"

"Given $AC \cong DF$, then for any point B between A and C, there is a unique point E between D and F such that $AB \cong DE$."

(pg 133) "Proposition 3.13b"

"If $AB < CD$ and $CD \cong EF$, then $AB < EF$"

Proof:

Hypothesis	Evidence
1) A point G appears on segment CD, such that $AB \cong CG$	Proposition 3.12
2) A point H appears on EF, such that $AB \cong EH$	Proposition 3.12
3) $GD \cong HF$	Proposition 3.11

(pg 133) "Proposition 3.13c"

"If $AB > CD$ and $CD \cong EF$, then $AB > EF$"

Proof:

Hypothesis	Evidence
1) A point G appears	Proposition 3.12

Hypothesis	Evidence
on segment AB, such $AG \cong CD$	
2) $AG \cong EF$	Axiom C-2

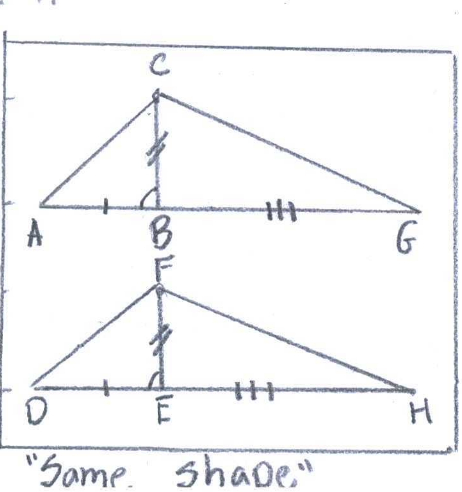
23. (pg 113) "Proposition 3.13d"

"If $AB < CD$ and $CD < EF$, then $AB < EF$."

Proof:

Hypothesis	Evidence
1) A point G resides on segment EF, such $CD \cong EG$	Proposition 3.12
2) Another point H resides on CD, such $AB \cong CH$	Proposition 3.12
3) $0 < HD < GF$	(pg 124) "Definition-Transitive" "Inequality"

24.



"Same shade."

(pg 125) "Proposition 3.14"

"Supplements of congruent angles are congruent."

Proof:

Hypothesis	Evidence
1) Triangle $\Delta ABC \cong \Delta DEF$	Axiom C-6

where $\measuredangle B = \measuredangle E$	
2) A unique ray $\overrightarrow{AB}$ is on side AB	Axiom C-4
3) A unique ray $\overrightarrow{DE}$ is on side DE	Axiom C-4
4) A point G is on AB and a point H is on DE	Axiom B-2
5) H is on DE for $BG \cong EH$	Proposition 3.11
6) $AG \cong DH$	Axiom C-3
7) $\triangle ACG \cong \triangle DFH$	Axiom C-6 (SAS)
8) $\triangle CBG \cong \triangle FEH$	Axiom C-6 (SAS)
9) $\measuredangle CBG \cong \measuredangle FEH$	Pon's Asinorum

25. (pg125) "Proposition 3.15"

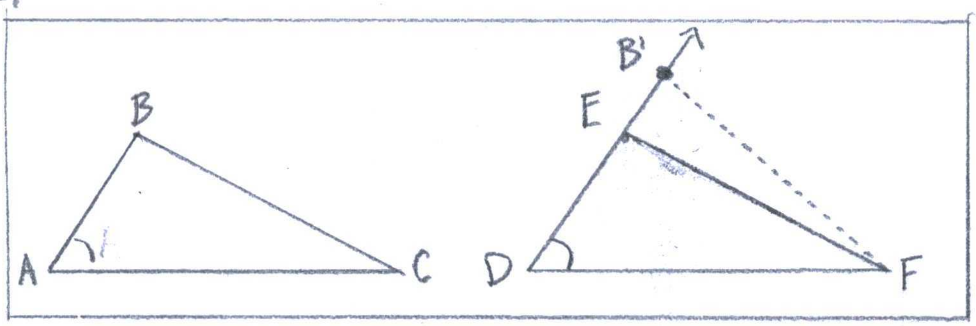
a) Vertical angles are congruent to each other

b) An angle congruent to a right

angle are congruent to each other."

Euclid deduced in the fourth (IV) postulate congruent right angles. Specially, Proposition 3.14 justified right angles by the supplementary counterpart.

26.



"ASA Criterion for Congruence"

(pg 127) "Proposition 3.17"

"Given ΔABC and ΔDEF with $\measuredangle A \cong \measuredangle D$, $\measuredangle C \cong \measuredangle F$, and $AC \cong DF$. Then $\Delta ABC \cong \Delta DEF$."

Proof: Hypothesis	Evidence
1) There is a unique point B' on ray $\overrightarrow{DE}$ such that $DB' \cong AB$	Axiom C-1

2) $\Delta ABC \cong \Delta DB'F$	$DB' \cong AB$, $\measuredangle A \cong \measuredangle D$ $AC \cong PF$ is Axiom C-6
3) $\measuredangle DFB' \cong \measuredangle C$	Axiom C-5
4) This implies $\overrightarrow{FE} = \overrightarrow{FB'}$	RAA Hypothesis
5) In that case, $B' = E$	Contradiction
6) Hence, $\Delta ABC \cong \Delta DEF$	

27. (pg 127) "Proposition 3.18"

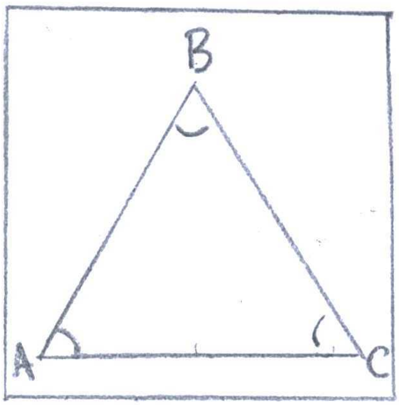
"If in ΔABC we have $\measuredangle B \cong \measuredangle C$, then $AB \cong AC$ and ΔABC is isoscles"

Proof:

Hypothesis	Evidence
1) There is a unique point B' on ray DE such $DB' \cong AB$	Axiom C-1
2) $\Delta ABC \cong \Delta DB'F$	$DB' \cong AB$, $\measuredangle A \cong \measuredangle D$, $AC \cong DF$, Axiom C-6
3) $\measuredangle DFB' \cong \measuredangle C$	Axiom C-5
4) This implies $\overrightarrow{FE} = \overrightarrow{FB'}$	RAA Hypothesis
5) $B' = E$	Contradiction

6) Hence, $\triangle ACD \cong \triangle BCD$	Axiom C-4
7) Therefore, $\measuredangle B \cong \measuredangle C$	Proposition 3.10
8) $\triangle ABC$ is isosceles	(pg 31) I. 5

28.

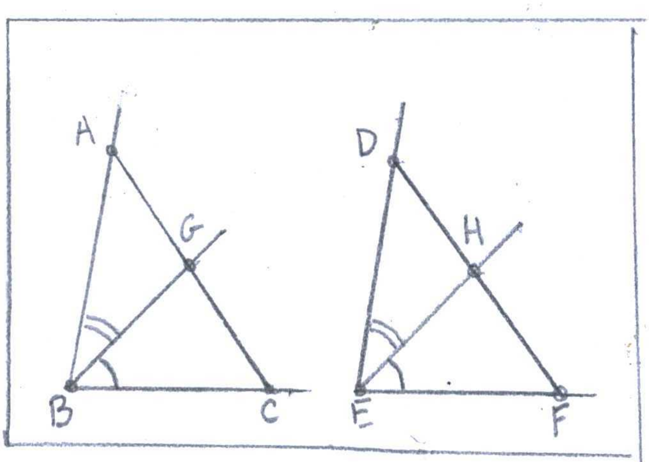


"Equiangular Triangle"

An Equiangular Triangle is Equilateral

Proof: Hypothesis	Evidence
1) $\measuredangle B \cong \measuredangle C$, $\measuredangle A \cong \measuredangle B$, $\measuredangle A \cong \measuredangle C$	Conjecture to Equiangular
2) $AB \cong AC$	Converse to Proposition 3.10
3) $\triangle ABC \cong \triangle ACB$	Implication by RAA
4) $AB \cong AC$, $BC \cong AC$, $AB \cong BC$	Equilateral

29.



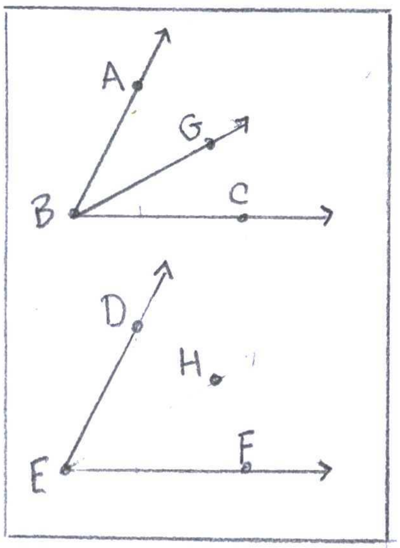
(pg 127) "Proposition 3.20 (Angle Subtraction)"

"Given $\overrightarrow{BG}$ between BA and BC, EH between ED, EF, $\measuredangle CBG \cong \measuredangle FEH$, and $\measuredangle ABC \cong \measuredangle DEF$ Then $\measuredangle GBA \cong \measuredangle HED$."

Proof:

Hypothesis	Evidence
1) $\measuredangle ABC \cong \measuredangle DEF$	Converse Proposition 3.19
2) $\measuredangle GBA \cong \measuredangle HED$	Axiom C-4

30.



$\measuredangle ABC \cong \measuredangle DEF$ and $\overrightarrow{BG}$ between $\overrightarrow{BA}$ and $\overrightarrow{BC}$ implies $\overrightarrow{EH}$ between $\overrightarrow{ED}$ and $\overrightarrow{EF}$:

Proof

Hypothesis	Evidence
1) A point H is between ED and EF	Proposition 3.7
2) $AG \cong DH$	Proposition 3.12
3) $AB \cong DE$	Axiom C-1
4) $\measuredangle CBG \cong \measuredangle FEH$	Contrapositive to Proposition 3.19
5) $\measuredangle ABC \cong \measuredangle DEF$	False contrapositive
6) $\measuredangle ABG \cong \measuredangle DEH$	Proposition 3.20

31. (pg 128) "Proposition 3.21 (Ordering of Angles)"

a) Exactly one of the following three conditions (trichotomy): $\measuredangle P < \measuredangle Q$, $\measuredangle P \cong \measuredangle Q$, or $\measuredangle Q < \measuredangle P$

b) If $\measuredangle P < \measuredangle Q$ and $\measuredangle Q \cong \measuredangle R$, then $\measuredangle P < \measuredangle R$

Case 5:

1) A ray $\vec{p}$ is between $\measuredangle P$, so $\measuredangle p \cong \measuredangle R$	(pg 126) "Definition - Angle Inequality"

Case 6:

1) A ray $\vec{r}$ is between $\measuredangle R$, so $\measuredangle P \cong \measuredangle r$	(pg 126) "Definition - Angle Inequality"

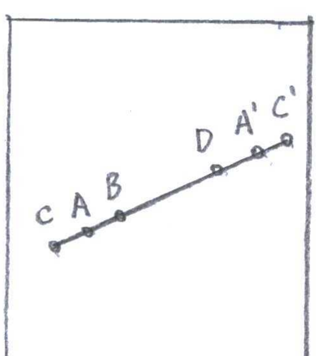
32. (pg 128) "Proposition 3.22 (SSS criterion for congruence)

"If $AB \cong DE$, $BC \cong EF$, and $AC \cong DF$, then $\Delta ABC \cong \Delta DEF$"

Proof:

Hypothesis	Evidence
1) Point F is on DE so $\Delta ABC \cong \Delta DEF$	Corollary to Axiom C-6
2) $\measuredangle A \cong \measuredangle D$ and $\measuredangle C \cong \measuredangle F$	"Proposition 3.17 (ASA Criterion for congruence)"

33. (From exercise)



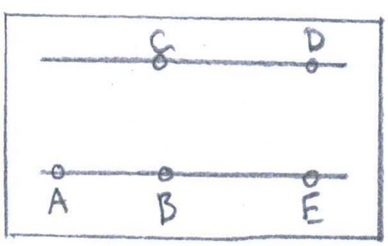
"If $AB < CD$, then $2AB < 2CD$"

Proof:

Hypothesis	Evidence.
1) Points A, B, C, D, A', and C' are on a line	Proposition 3.3

2) Point $B = D$	Conjecture to an RAA
3) $AB \cong BA'$	Axiom C-1
4) $CB \cong BC'$	Axiom C-1
5) $AA' < CA'$	Proposition 3.21
6) $CA' < CC'$	Proposition 3.21
7) $AA' \cong 2AB$ and $CC' = 2BC$	(pg 65) "Logic Rule-9 $P \to Q \to R$
8) $2AB < 2BC$	
9) But point B is not = D, so $2AB < 2CD$	Contradiction

34.



"Euclid's Postulate II"

a) (pg 16) "Euclid's Postulate II"

"For every segment AB and for every segment CD there exists a unique point E on line AB such that B is E is between A and B and segment CD is congruent to segment BE."

Proof:

Hypothesis	Evidence
1) $CD \cong BE$	Axiom C-1
2) Another point A is on BE for $A * B * E$.	Axiom B-2

b) (from exercise)

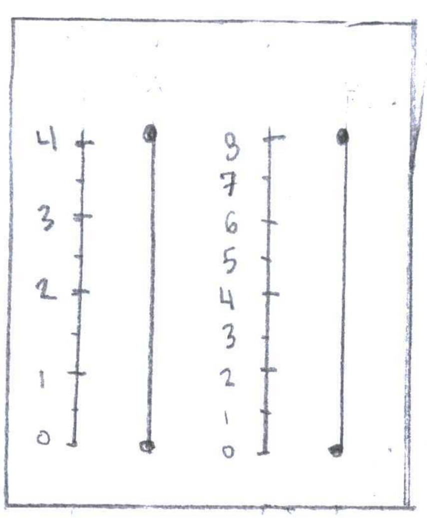
"The center of a circle is unique and its radius is unique up to congruence"

Proof:

Hypothesis	Evidence
1) Points O and O' exist with radius OA and OA'	Euclid's Postulate III
2) Circles are unique unless O=O' and OA=OA'	(Circle O)$\cap$(Circle O')

35.

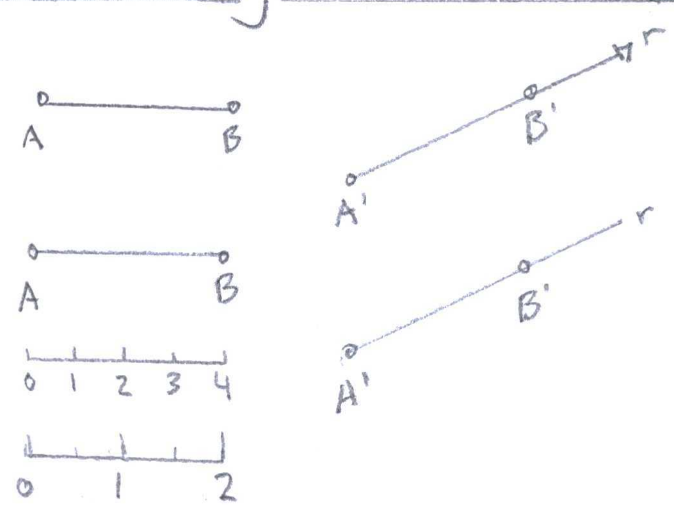
(From exercise)



"Two segments of the same length have the same length in this perverse way of measuring."

Informal proofs describing Axioms C-1/2/3/4/5

Axiom C-1:



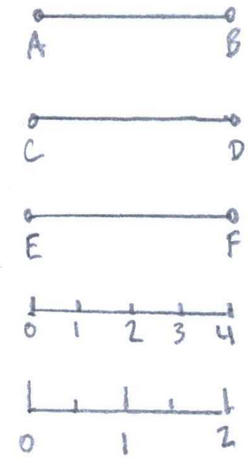
c) If $\measuredangle P > \measuredangle Q$ and $\measuredangle Q \cong \measuredangle R$, then $\measuredangle P > \measuredangle R$

d) If $\measuredangle P < \measuredangle Q$ and $\measuredangle Q < \measuredangle R$, then $\measuredangle P < \measuredangle R$

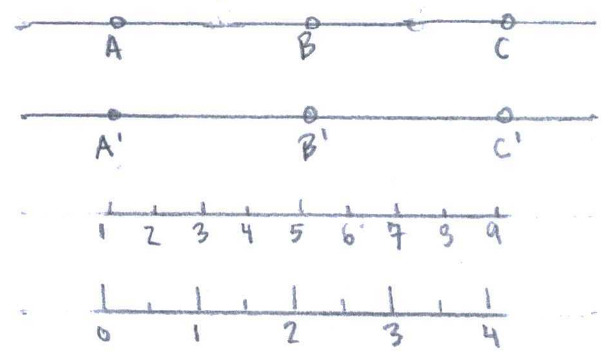
Proof:

Hypothesis	Evidence
Case 1:	
1) An angle $\measuredangle R$ where $\measuredangle R \cong \measuredangle P$ and $\measuredangle R \cong \measuredangle Q$	Axiom C-4
2) $\measuredangle P \cong \measuredangle Q$	
Case 2:	
1) An angle $\measuredangle R$ where $\measuredangle R \cong \measuredangle P$ and $\measuredangle R < \measuredangle Q$	Axiom C-4
2) $\measuredangle P < \measuredangle Q$	
Case 3:	
1) An angle $\measuredangle R$ where $\measuredangle R \cong \measuredangle P$ and $\measuredangle R > \measuredangle Q$	Axiom C-4
2) $\measuredangle P > \measuredangle Q$	
Case 4:	
1) A ray $\vec{r}$ is in $\measuredangle R$, so $\measuredangle P \cong \measuredangle r$	(pg 128) "Definition - Angle Inequalities."

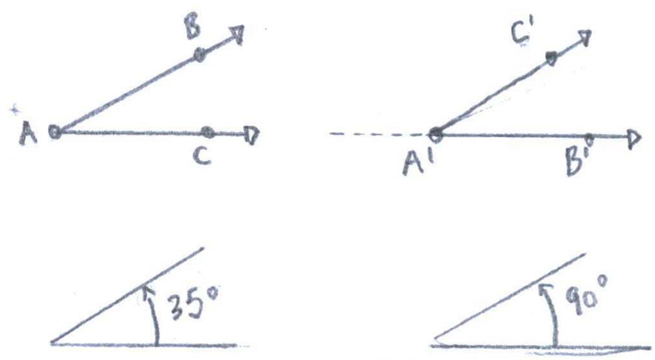
Axiom C-2:



Axiom C-3:

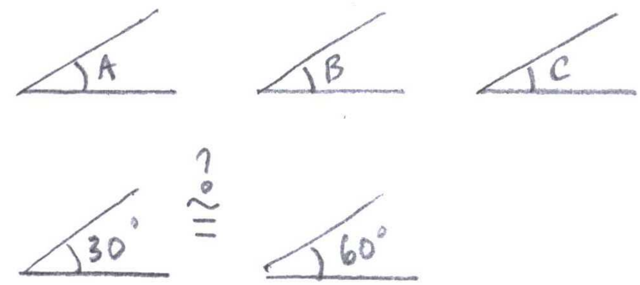


Axiom C-4:

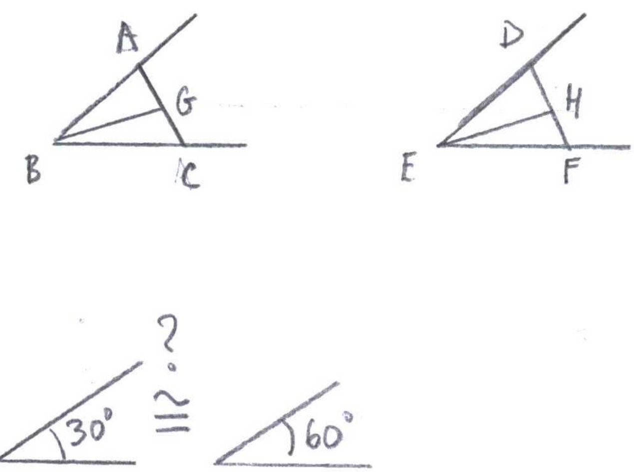


Note: Axiom C-4 may be false to mathematicions with "perverse" angles. The total degrees is constant around a circle from radians relationship to degrees. Base-60 describes Pi (π) in a series.

Axiom C-5:



Proposition 3.19:



Axiom C-6 Fails with all triangles because prerequisites, including sum total angles.

Circle-circle continuity principle fails because 360° notation.

Segment-circle principle cannot follow the "perverse" measurement because absolute degrees inside a circle.